

UniProcure Institutional Portal

University E-Procurement & Vendor Management System — Frontend
Architecture, Capabilities, Security & Strategic Roadmap

TARGET SECTOR

Higher Ed & Public
Research

FRONTEND TECH

Flutter Web 3.x / Dart 3.x

COMPLIANCE LEVEL

PPA/PPR Tender
Regulations

DOCUMENT CLASS

Client Commercial Pitch

1. Executive Summary & Commercial Value Proposition

UniProcure is a high-security, web-native enterprise procurement management portal tailored to address the rigorous statutory governance, multi-tiered fiscal approvals, and competitive bidding workflows of universities and public academic bodies.

Key Institutional Value Proposition for Decision Makers:

- **Statutory Procurement Integrity:** Built specifically to comply with public procurement guidelines (e.g. mandatory 3-quote rule, transparent tender publication periods, sealed competitive quotation evaluations).
- **Fraud & Leakage Prevention:** Eliminates manual paper files and informal back-channel approvals by establishing rigid audit trails with immutable officer logs.
- **Objective Bid Evaluation:** Built-in *Quality & Cost-Based Selection (QCBS)* automated comparison matrix scoring technical compliance alongside financial bids.
- **Zero-Backend Presentation Ready:** Features an integrated client-side reactive simulation engine (`DummyDatabaseService`) that permits live, fully interactive demonstrations to evaluation committees and board members with zero server prerequisites.

2. Frontend Software Technologies & Architecture

The client application is engineered with an enterprise Flutter Web stack compiled for desktop-class web performance:

Technology	Package / Tool	Role & Architectural Rationale
Core Client Engine	Flutter Web (Dart 3.12+)	Hardware-accelerated web rendering delivering responsive, pixel-perfect visual consistency across all modern enterprise desktop and mobile browsers.
Architecture Pattern	Clean Architecture + DDD	Decoupled presentation, domain (use cases, entities), and data layers to isolate business logic from UI widgets and backend communications.
State & Dependency DI	GetX (^4.7.2)	Ultra-responsive micro-state management (Rx , Obx), declarative dependency injection (GetxService), and typed named route navigation.
Network Engine	Dio (^5.7.0)	Robust HTTP client configured with automated token injection, custom interceptors, connection pooling, and timeout guards.
Fault Resiliency	RetryInterceptor	Automatic HTTP retry mechanism with exponential backoff and jitter for unstable campus networks and high-load periods.
Client Storage	GetStorage (^2.1.1)	Synchronous, fast key-value storage abstracted over HTML5 localStorage , eliminating native mobile plugin crashes.
Error Safety Monads	Dartz (^0.10.1)	Functional error handling using Either<Failure, T> monads, eliminating uncaught frontend exceptions.
Responsive Data Tables	DataTable2 (^2.5.15)	High-density data grids with pinned headers, horizontal scroll, column sorting, and custom status cell renderers.
Visual Analytics	FL Chart (^1.2.0)	Hardware-accelerated charting for fiscal budget utilization, departmental spend distribution, and 6-month trend analysis.
Edge Cloud Deployment	Vercel Edge / CDN	Automated CI/CD build scripts (vercel-build.sh) with SPA rewrite rules (vercel.json) for instantaneous global caching.

3. UI/UX Design System & Responsive Layout

- Institutional Design System:** Professional slate-and-navy aesthetic (Primary Blue #1A56DB , Deep Navy #0F172A , Emerald #16A34A , Amber #F59E0B) custom-styled for institutional software.
- Three-Tier Adaptive Layout Engine:**
 - Desktop (≥ 1200px):** Permanent 260px branded sidebar with live badges, dual-column analytics, and full data grids.
 - Tablet (600px – 1199px):** Collapsible 72px icon rail that expands on hover to 220px, optimizing content viewport.
 - Mobile (< 600px):** Elevation-zero AppBar with slide-out navigation drawer and single-column action cards.

- **RBAC Dynamic Menus:** The sidebar navigation adapts dynamically via `PermissionService`, displaying only the modules pertinent to the logged-in officer.

4. In-Depth Features & Functionalities by Persona

UniProcure isolates capabilities across five authenticated roles plus a public access tier:

1. Institutional Admin

GOVERNANCE

- **User Management:** Activate/deactivate faculty accounts, reassign roles, view pending workload count.
- **Vendor Verification Queue:** Review statutory TIN, Trade License, and VAT certificates before clearing suppliers.
- **Workflow Hierarchy Engine:** Configure monetary approval limits (Tier 1 Dept Head, Tier 2 Dean, Tier 3 Registrar).
- **Governance Rules:** Enforce 3 minimum bids, strict budget caps, and auto-escalation timer windows.
- **Executive Analytics:** 8-metric KPI suite, spend by category, 6-month trends, top vendor scorecard.

2. Requisition Initiator

DEPARTMENT

- **Circular Creation:** Author procurement notices with itemized Bill of Quantities (BOQ) and budget limits.
- **Specification Upload:** Attach technical drawings, terms of reference (TOR), and procurement rules.
- **Tender Lifecycle Cockpit:** Monitor live bidding radar (bid counts, days remaining, submission status).
- **Project History Archive:** Complete audit record of fulfilled purchase orders and delivery certificates.

3. Multi-Tiered Approvers

SANCTIONING

- **Tiered Queue:** Dedicated queues for Department Head, Faculty Dean, and Registrar / VC.
- **Side-by-Side Review:** Inspect comparative vendor quotations, technical scores, and budget deltas.
- **Formal Decision Cockpit:** Approve with digital remarks, reject with mandatory justification, or return for revision.
- **Approval Audit Tracker:** Visual chronological stepper displaying officer timestamps and comments.

4. Registered Vendors

CONTRACTORS

- **Digital Onboarding:** Company profile registration with statutory document attachment.
- **Public Tender Explorer:** Browse, filter, and review open institutional tender notices.
- **Online Bid Submission:** Submit quoted price, delivery timeline, compliance certification, and PDF dossiers.
- **'My Bids' Tracker:** Real-time status indicators (Submitted, Evaluated, Awarded, Disqualified).
- **Work Orders & Invoicing:** Milestone delivery reporting and commercial tax invoice uploads.

5. Finance & Treasury Officers (Controllership)

- **Fiscal Budget Monitoring:** Real-time monitoring of Annual Budget (\$3.5M baseline), Committed Liabilities (\$2.18M), and Disbursed Funds (\$1.42M).
- **3-Way Matching Verification:** Digital cross-verification between Awarded Work Order, Lab Goods Receipt Note, and Vendor Commercial Invoice.
- **Direct EFT Disbursal Authorization:** Settle verified claims, generate Electronic Funds Transfer (EFT) batch references, or return invoices for clarification.
- **Departmental Spend Allocation:** Track fund absorption across Science, Engineering, Business, and Arts faculties.

Signature Capability: Automated QCBS Bid Comparison Matrix

UniProcure features an institutional evaluation matrix implementing **Quality & Cost-Based Selection (QCBS)**:

- **Composite Score Calculation:** Combines technical evaluation scores (e.g. 70% weight) with commercial quotation competitiveness (e.g. 30% weight).
- **Budget Delta Analysis:** Instantly displays quotation variance relative to the approved ceiling (under-budget vs. budget overrun).
- **Automated Award Recommendation:** Formulates institutional committee recommendation for immediate executive sanctioning.

5. Production Readiness & Web Security Architecture for Flutter Web

Deploying Flutter Web in enterprise and governmental settings mandates strict client-side hardening and edge infrastructure configurations:

Critical Production Requirements for Web Publishing:

- **Rendering Engine Selection:** Build using Flutter's `CanvasKit` engine (`flutter build web --release --no-wasm-dry-run`) to guarantee pixel-identical charts, typography, and complex data grids across all browsers.
- **Content Security Policy (CSP):** Configure edge servers with strict CSP headers permitting WebAssembly execution (`'wasm-unsafe-eval'`) while blocking unauthorized script origins.
- **Cache Eviction & Service Worker:** Utilize the cache-clearing bootstrap in `web/index.html` to ensure users immediately receive updated bundles upon deployment without manual hard-refreshing.
- **Session Security & Token Shielding:** Employ in-memory access token storage backed by HTTP-Only, Secure, SameSite=Strict cookies for token refresh cycles to neutralize XSS token harvesting.
- **Client Route Guards:** `AuthGuard` and `PermissionService` validate permissions before each route transition, preventing unauthorized direct URL navigation.

Recommended Production Content Security Policy (CSP) Header:

```
Content-Security-Policy: default-src 'self';  
script-src 'self' 'wasm-unsafe-eval' https://fonts.googleapis.com;  
style-src 'self' 'unsafe-inline' https://fonts.googleapis.com;  
font-src 'self' https://fonts.gstatic.com;  
img-src 'self' data: blob: https;;  
connect-src 'self' https://api.uniprocure.edu;  
object-src 'none'; frame-ancestors 'none';
```

6. Strategic Product Roadmap (Future Phases to Close the Deal)

The following realistic, high-impact features represent the immediate product roadmap to present to university leadership and expansion partners:

Phase / Target	Feature Module	Commercial Impact & Operational Capability
Phase 1: Legal Compliance	PKI Digital Signatures	Integrate cryptographic X.509 digital certificates and enterprise e-signatures (DocuSign / National PKI) to legally seal approved sanction orders and tender notices.
Phase 2: Live Collaboration	Real-Time WebSockets	Instant notifications for tender closing countdowns, outbid supplier alerts, and executive approval pings without page refresh.
Phase 3: Audit Automation	In-Browser PDF Generator	Client-side compilation and download of formatted official Purchase Orders, Comparative Evaluation Statements, and Treasury Ledgers.
Phase 4: Fiscal Governance	Automated Tax & VAT Calculator	Dynamic calculation of Advanced Income Tax (AIT) and Value Added Tax (VAT) deductions at source during invoice audit prior to bank clearance.
Phase 5: Risk Mitigation	Vendor Performance Scorecard	Automated vendor rating factoring delivery delays and SLA penalties, dynamically weighted in future tender evaluations.
Phase 6: Scalability	Multi-Campus Federation	Multi-tenant organization switcher supporting affiliated medical colleges, specialized research centers, and satellite campuses under one portal.

7. Summary of Deliverables & Demonstration Verification

8	25+	100%
Functional Modules	Screen Views & Pages	Responsive & Web-Ready

Build Command:

```
flutter build web --release --no-wasm-dry-run
```